REZON fall detector · wiring list

SM-24 geophone · ADC 15 Click · ESP32-S3 · PCM5102A + PAM8403 · SIM868 (Waveshare) · 18650 · hackathon build · 2026-11-12 · same numbering as the 3D guide (Rezon_Wiring_Guide.html)

HOW TO READ A HOLE

- Main board: **rows A … Z, AA** (27) from the back edge to the front edge; **columns 1 … 35** from left to right, **seen from the component side** with the rails and H1 at the front.
- **K6** = row K, column 6. Lid board: rows A–H, columns 1–12, same rule.
- On the **solder side** (board turned over like a page) row A stays at the top and **column 1 is on the right**. All maps of solder-side work are drawn that way.
- Write “A” and “1” on both sides of the board before you start.

HOW EVERY CONNECTION IS MADE

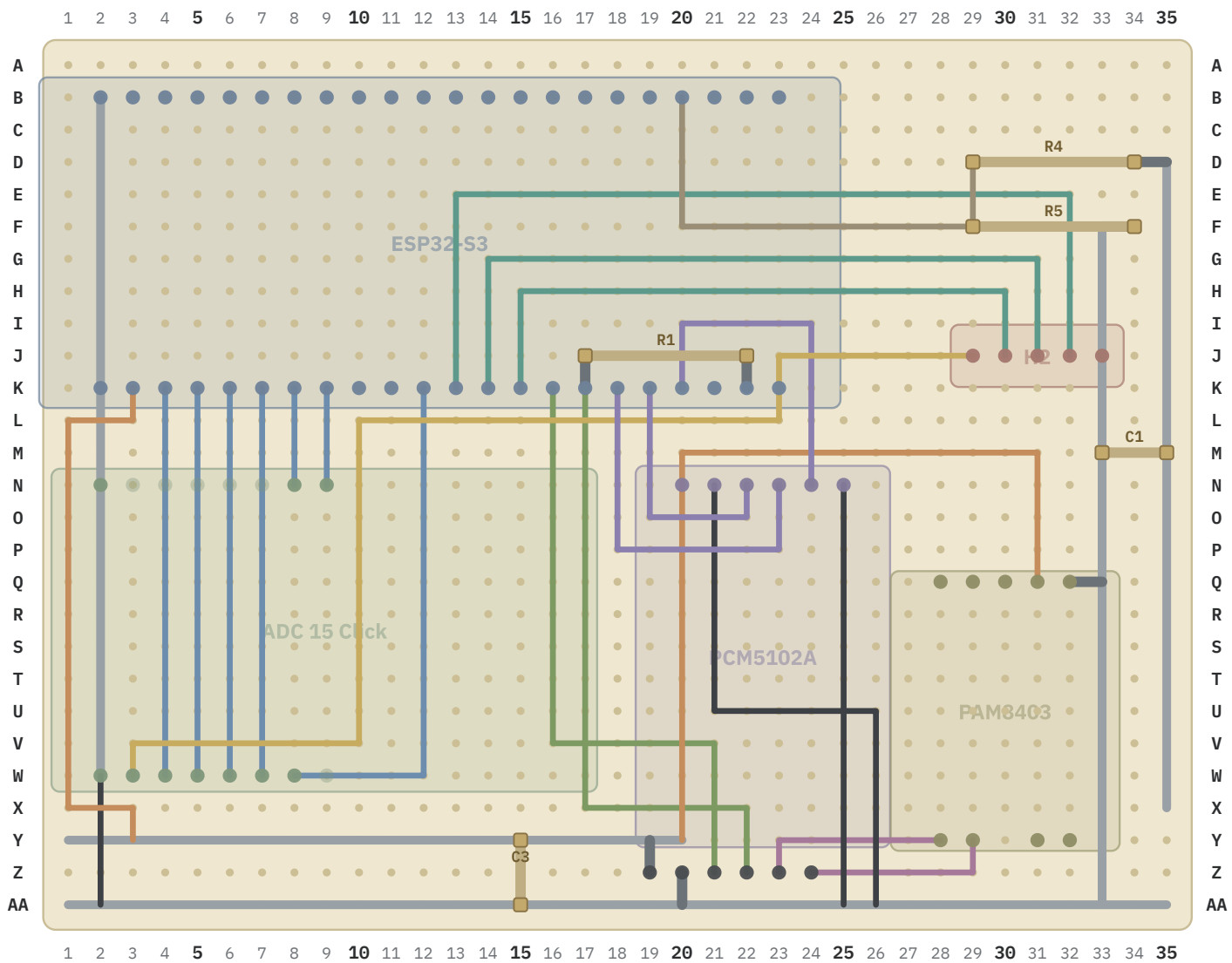
- **Bare wire** (rails/buses): 20 AWG solid, laid across the pads on the solder side, soldered at each listed hole.
- **Bridge**: bend the component leg flat onto the neighbouring pin and solder both (one joint).
- **Link** (33×): one insulated core from the 6×0.22 cable on the solder side. Cut to the listed length, strip 3 mm both ends, solder each end onto the named pin leg. “via” lists the corner holes the wire passes over.
- **Top wire**: on the component side (screw terminals, module pads).
- Two wires may cross (insulated); a link never runs along a bare rail.

COLOURS USED IN THIS LIST

GND · 5 V · VBAT · 3V3 · SPI · I²S · UART · Speaker · LED/button · Battery sense · Bare wire

Colours are for reading the list. The insulated cores can be any colour; label both ends with tape if they are all the same.

BOARD LAYOUT · COMPONENT SIDE



COMPONENT SIDE · row A (back) at the top · column 1 on the left · solder-side wires shown through the board

Check your modules first. The ESP32-S3 rows must be 9 holes apart (B and K), the Click rows 9 apart (N and W). If one of yours differs, shift that module's second row and every hole that refers to it. Firmware pin map: SCK 12, MISO 13, MOSI 14, ADC-CS 11, clock-CS 9, DRDY 10, ADC-RST 8, I²S BCK 4 / LCK 5 / DIN 6, SIM868 TXD→GPIO18, GPIO17→SIM868 RXD, SIM868 PWRKEY 16 (HIGH 1.5 s), button 7, LED 15, VBAT/2 on GPIO1.

REF Every pin hole on the main board

Use this to check a hole before soldering. Pins are listed per module, left to right on the component side.

ESP32-S3

B2	J3-22 GND
B3	J3-21 GND
B4	J3-20 GPIO19
B5	J3-19 GPIO20
B6	J3-18 GPIO21
B7	J3-17 GPIO47
B8	J3-16 GPIO48
B9	J3-15 GPIO45
B10	J3-14 GPIO0
B11	J3-13 GPIO35
B12	J3-12 GPIO36
B13	J3-11 GPIO37
B14	J3-10 GPIO38
B15	J3-9 GPIO39
B16	J3-8 GPIO40
B17	J3-7 GPIO41
B18	J3-6 GPIO42
B19	J3-5 GPIO2
B20	J3-4 GPIO1
B21	J3-3 RX
B22	J3-2 TX
B23	J3-1 GND
K2	J1-22 GND
K3	J1-21 5V
K4	J1-20 GPIO14
K5	J1-19 GPIO13
K6	J1-18 GPIO12
K7	J1-17 GPIO11
K8	J1-16 GPIO10
K9	J1-15 GPIO9
K10	J1-14 GPIO46
K11	J1-13 GPIO3
K12	J1-12 GPIO8
K13	J1-11 GPIO18
K14	J1-10 GPIO17
K15	J1-9 GPIO16
K16	J1-8 GPIO15
K17	J1-7 GPIO7
K18	J1-6 GPIO6
K19	J1-5 GPIO5
K20	J1-4 GPIO4
K21	J1-3 EN
K22	J1-2 3V3
K23	J1-1 3V3

CLICK

N2	GND
N3	NC
N4	NC
N5	NC
N6	NC
N7	NC
N8	INT (DRDY)
N9	CS2
W2	GND
W3	3V3
W4	SDI (MOSI)
W5	SDO (MISO)
W6	SCK
W7	CS1
W8	RST
W9	NC

PCM5102A

N20	VIN
N21	GND
N22	LCK
N23	DIN
N24	BCK
N25	SCK

H2

J29	3V3 → SIM868 pin 36
J30	PWRKEY → SIM868 pin 19
J31	SIM RXD → SIM868 pin 1
J32	SIM TXD → SIM868 pin 2
J33	GND → SIM868 pin 3

PAM8403

Q28	L in
Q29	⏚ in
Q30	R in
Q31	+5V
Q32	GND
Y28	L+ out
Y29	L– out
Y31	R– out
Y32	R+ out

H1

Z19	5V
Z20	GND
Z21	LED
Z22	BTN
Z23	SPK+
Z24	SPK–

PARTS ON THE MAIN BOARD

C3	470 µF · Y15 +, AA15 – · step 4
C1	1000 µF · M35 +, M33 – · step 8
R1	10 kΩ · J17, J22 · step 9
R4	100 kΩ · D34, D29 · step 9
R5	100 kΩ · F29, F34 · step 9

BARE RAILS AND BUSES

AA1–AA35	GND rail
Y1–Y20	5 V rail
D35–X35	VBAT rail (switched battery)
B2–W2	GND bus col 2
F33–AA33	GND bus col 33

OFF-BOARD WIRES LAND IN

X35	VBAT from rocker pin 2
W35	VBAT to boost IN+
AA35	GND from TP4056 OUT–
AA34	GND to boost IN–
Y1	5 V from boost OUT+
AA1	GND from boost OUT–
AA18 AA19	GND wires up to the Click screw terminals
L35	VBAT to the SIM868 battery plug (red)
L33	GND to the SIM868 battery plug (black)

STEP 2 Cut the perfboard

Cutting bench

Cut the 9 × 15 cm perfboard (stock row 101) into the main board (35 columns × 27 rows of holes, ≈ 90 × 70 mm) and the lid board (12 × 8 holes, ≈ 30 × 20 mm). Score both sides along a hole row, snap, sand flat, knock 1 mm off each corner. With a pen, write “A” at the back edge and “1” at the left edge on BOTH sides of the main board.

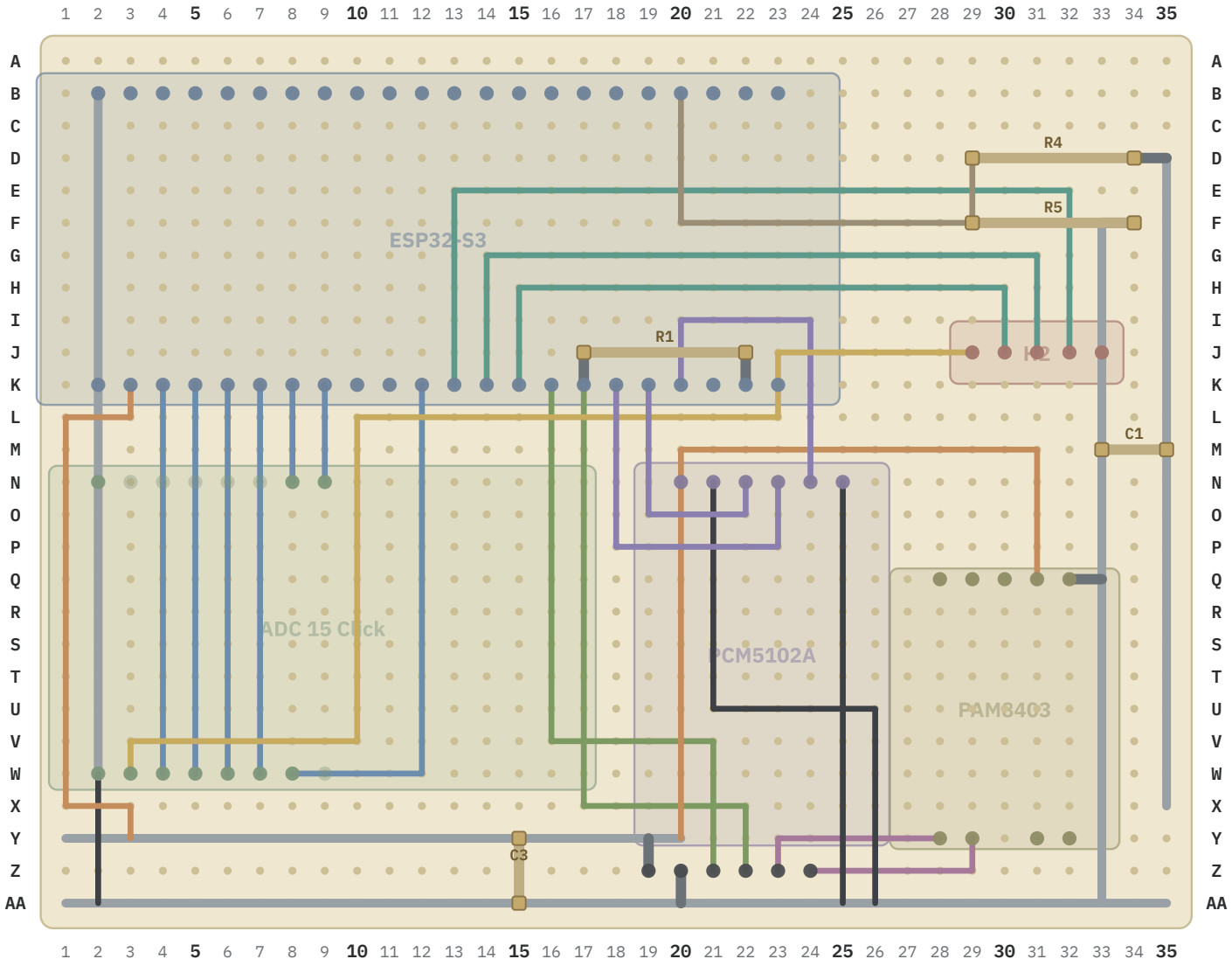
STEP 3 Sockets, headers & module pins

Soldering station · Cutting bench

Component side up. Solder every socket and header before any wire. Plug the module in while soldering so the rows stay parallel; tack one pin at each end first, check, then solder the rest.

HOLES

B2 … B23	ESP32-S3 J3 socket, 22 contacts. B23 = J3-1 (GND, antenna end), B2 = J3-22 (GND, USB end)
K2 … K23	ESP32-S3 J1 socket, 22 contacts. K23 = J1-1 (3V3), K3 = J1-21 (5V), K2 = J1-22 (GND)
N2 … N9	Click back header socket, 8 contacts. N9 = CS2, N8 = INT, N7–N3 = NC, N2 = GND
W2 … W9	Click front header socket, 8 contacts. W9 = NC, W8 = RST, W7 = CS1, W6 = SCK, W5 = SDO, W4 = SDI, W3 = 3V3, W2 = GND
J29 … J33	H2 SIM868 header, 5 male pins pointing up: J29 3V3, J30 PWRKEY, J31 SIM RXD, J32 SIM TXD, J33 GND (the SIM868 board itself is off the perfboard, at the back wall)
N20 … N25	PCM5102A: its own male header soldered straight into the board. N20 VIN, N21 GND, N22 LCK, N23 DIN, N24 BCK, N25 SCK
Q28 … Q32	PAM8403 input/power pins (male header pins, module 2 mm above the board): Q28 L in, Q29 ⏚ in, Q30 R in, Q31 +5V, Q32 GND
Y28 Y29 Y31 Y32	PAM8403 output pins: Y28 L+, Y29 L–, Y31 R–, Y32 R+
Z19 … Z24	H1 lid header, 6 male pins pointing up: Z19 5V, Z20 GND, Z21 LED, Z22 BTN, Z23 SPK+, Z24 SPK–



COMPONENT SIDE · row A (back) at the top · column 1 on the left · solder-side wires shown through the board

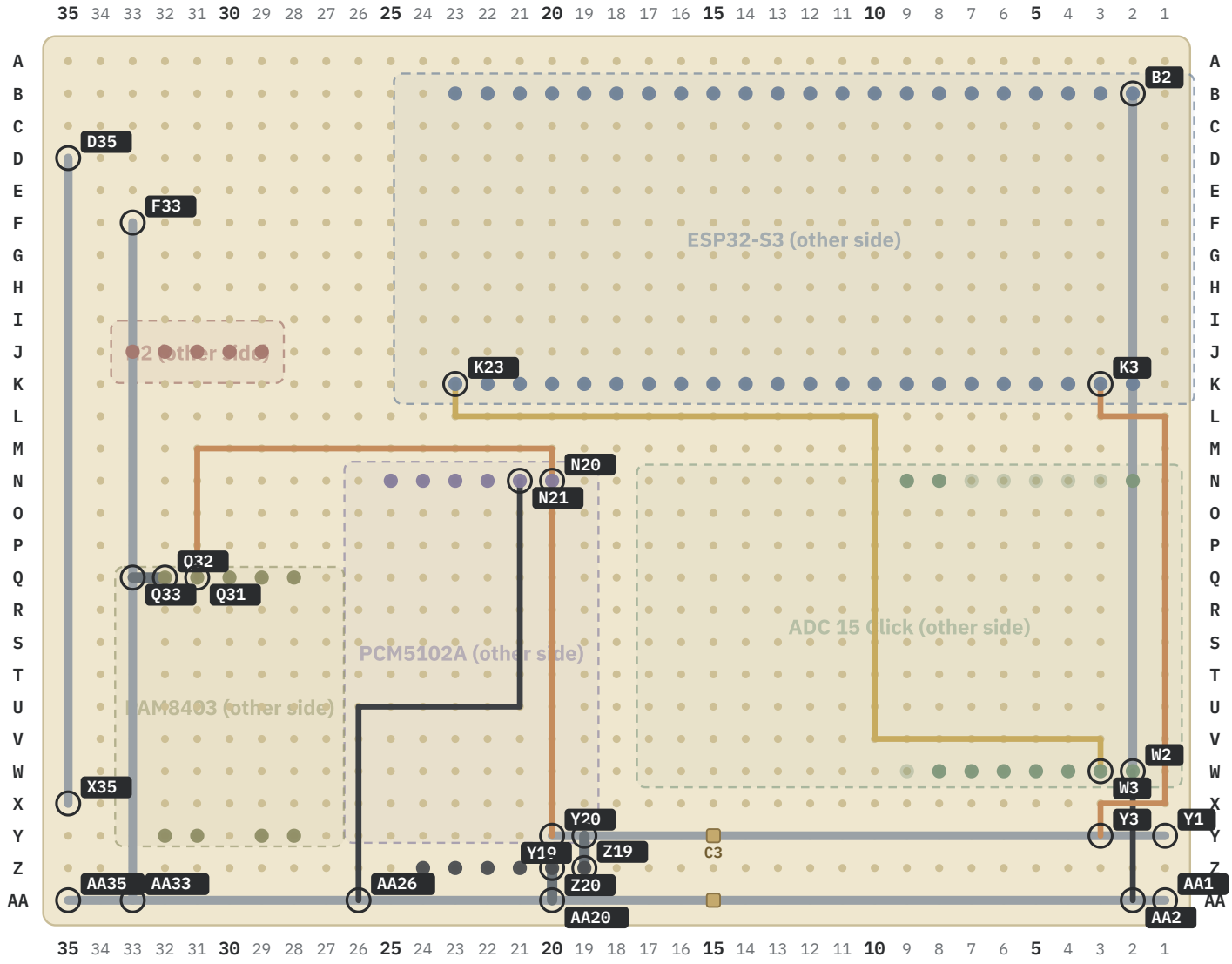
STEP 4 Rails, buses & power distribution

Soldering station · Cutting bench

Board turned over (solder side). Lay the bare rails first, then fit C3, then the bridges and links. Bare wire: straighten 20 AWG, lay it across the pads, solder at every hole that has a pin or leg and at every 4th hole in between.

DO THESE IN ORDER

#	TYPE	FROM	TO	ROUTE (SOLDER SIDE)	WIRE / CUT
4.1	Part	**C3** 470 µF · electrolytic, stripe = – legs into **Y15** + **AA15** – (component side)			bend legs flat after soldering
4.2	Bare wire	**AA1** **GND rail**	**AA35** **GND rail** GND rail (bare wire along row AA)	along AA1 → AA35 solder at: AA15, AA1, AA35	bare 20 AWG · cut 100 mm
4.3	Bare wire	**Y1** **5 V rail**	**Y20** **5 V rail** 5 V rail (bare wire along row Y)	along Y1 → Y20 solder at: Y15, Y1, Y20	bare 20 AWG · cut 60 mm
4.4	Bare wire	**B2** **ESP32-S3** J3-22 GND	**W2** **Click** GND GND bus col 2: ESP32 GND ↔ Click GND (bare)	along B2 → W2 solder at: B2, K2, N2, W2, B2, W2	bare 20 AWG · cut 65 mm
4.5	Bare wire	**F33** **GND bus** col 33	**AA33** **GND rail** GND bus col 33: H2 GND, C1–, SIM868 GND lead, PAM GND → rail (bare)	along F33 → AA33 solder at: J33, M33, F33, AA33	bare 20 AWG · cut 65 mm
4.6	Bare wire	**D35** **VBAT rail**	**X35** **VBAT rail** VBAT rail (bare wire along col 35)	along D35 → X35 solder at: M35, D35, X35	bare 20 AWG · cut 65 mm
4.7	Bridge	**Z19** **H1** 5V	**Y19** **5 V rail** H1 5V → 5 V rail		solder bridge / bent leg
4.8	Bridge	**Z20** **H1** GND	**AA20** **GND rail** H1 GND → GND rail		solder bridge / bent leg
4.9	Bridge	**Q32** **PAM8403** GND	**Q33** **GND bus** col 33 PAM8403 GND → GND bus		solder bridge / bent leg
4.10	Link	**W2** **Click** GND	**AA2** **GND rail** GND bus col 2 → GND rail	straight	insulated core · cut 20 mm, strip 3 mm both ends
4.11	Link	**Y3** **5 V rail**	**K3** **ESP32-S3** J1-21 5V 5 V rail → ESP32 5V	via X3 → X1 → L1 → L3	insulated core · cut 55 mm, strip 3 mm both ends
4.12	Link	**Y20** **5 V rail**	**N20** **PCM5102A** VIN 5 V rail → PCM VIN	straight	insulated core · cut 40 mm, strip 3 mm both ends
4.13	Link	**N20** **PCM5102A** VIN	**Q31** **PAM8403** +5V PCM VIN → PAM8403 +5V	via M20 → M31	insulated core · cut 50 mm, strip 3 mm both ends
4.14	Link	**N21** **PCM5102A** GND	**AA26** **GND rail** PCM GND → GND rail	via U21 → U26	insulated core · cut 55 mm, strip 3 mm both ends
4.15	Link	**K23** **ESP32-S3** J1-1 3V3	**W3** **Click** 3V3 ESP32 3V3 → Click 3V3	via L23 → L10 → V10 → V3	insulated core · cut 90 mm, strip 3 mm both ends



SOLDER SIDE · board turned over like a page · row A at the top · column 1 on the RIGHT

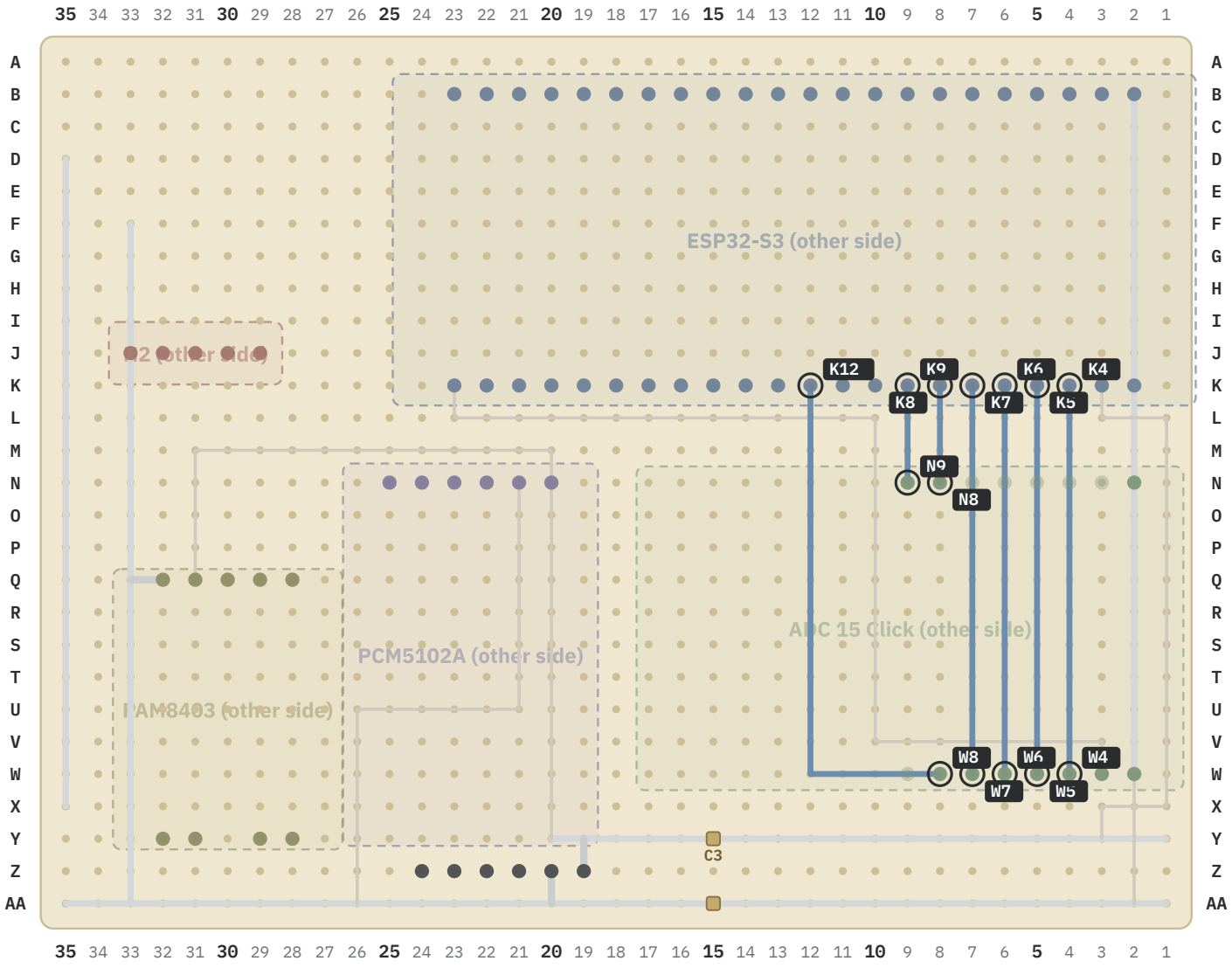
STEP 5 ADC 15 Click ↔ ESP32-S3 (SPI)

Soldering station

Seven links on the solder side. Six run straight in one column and pass the Click's unused NC pins (N3–N7, W9): keep the insulation intact there.

DO THESE IN ORDER

#	TYPE	FROM	TO	ROUTE (SOLDER SIDE)	WIRE / CUT
5.1	Link	K6 **ESP32-S3** J1-18 GPIO12	W6 **Click** SCK SCK	straight	insulated core · cut 40 mm, strip 3 mm both ends
5.2	Link	K5 **ESP32-S3** J1-19 GPIO13	W5 **Click** SDO (MISO) MISO	straight	insulated core · cut 40 mm, strip 3 mm both ends
5.3	Link	K4 **ESP32-S3** J1-20 GPIO14	W4 **Click** SDI (MOSI) MOSI	straight	insulated core · cut 40 mm, strip 3 mm both ends
5.4	Link	K7 **ESP32-S3** J1-17 GPIO11	W7 **Click** CS1 ADC chip-select	straight	insulated core · cut 40 mm, strip 3 mm both ends
5.5	Link	K9 **ESP32-S3** J1-15 GPIO9	N9 **Click** CS2 clock-chip select	straight	insulated core · cut 20 mm, strip 3 mm both ends
5.6	Link	K8 **ESP32-S3** J1-16 GPIO10	N8 **Click** INT (DRDY) DRDY, active low	straight	insulated core · cut 20 mm, strip 3 mm both ends
5.7	Link	K12 **ESP32-S3** J1-12 GPIO8	W8 **Click** RST ADC reset	via W12	insulated core · cut 50 mm, strip 3 mm both ends



SOLDER SIDE · board turned over like a page · row A at the top · column 1 on the RIGHT

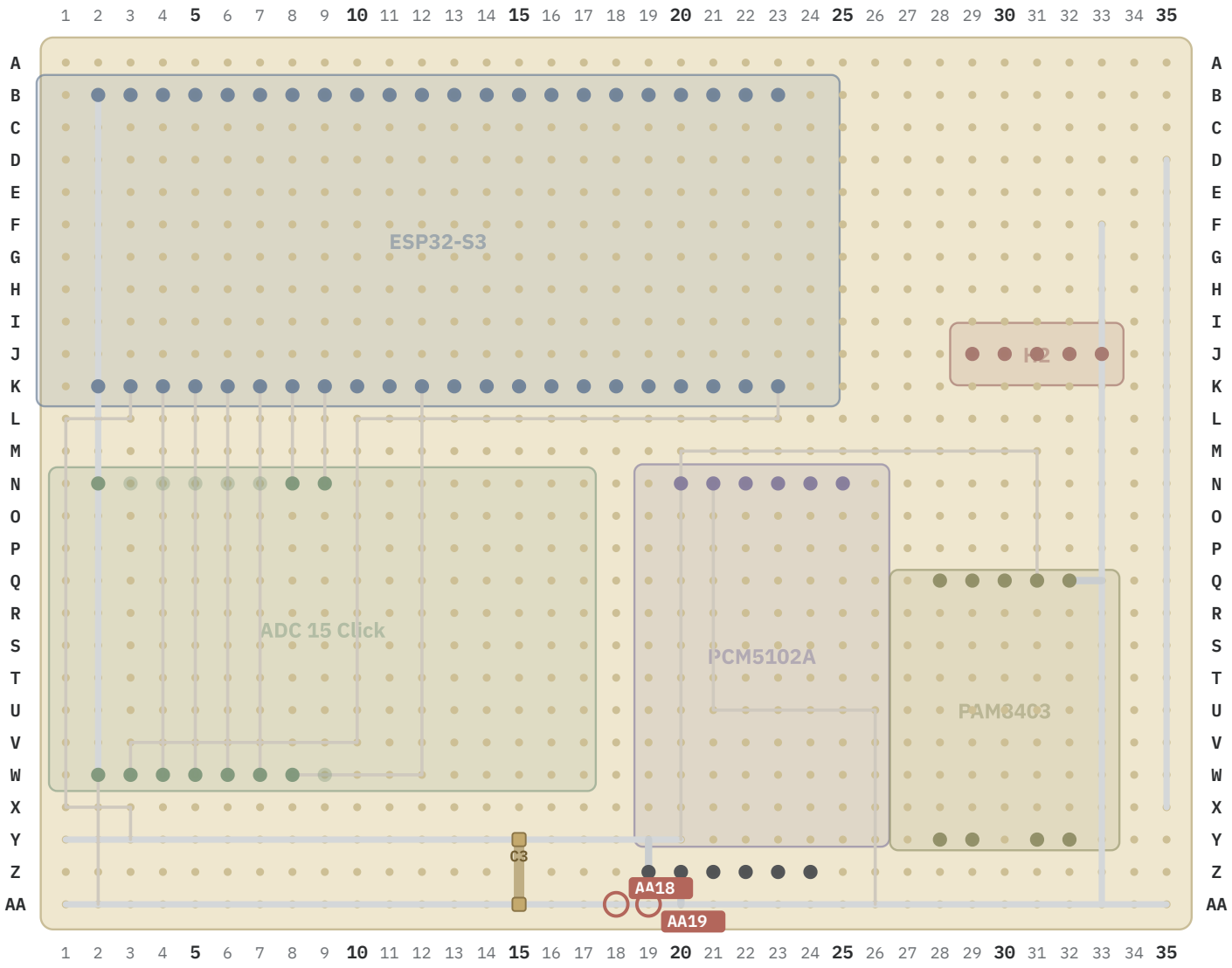
STEP 6 Geophone → Click CH1

Soldering station · Small flat screwdriver

Solder the 1 kΩ resistor across the two SM-24 solder tabs. Twist two cores of the 6×0.22 cable (about 1 turn per cm), solder to the tabs (mark the + core). Screw-terminal ends: strip 5 mm, twist, tin lightly. Each terminal takes at most two conductors.

DO THESE IN ORDER

#	TYPE	FROM	TO	ROUTE (SOLDER SIDE)	WIRE / CUT
6.1	Pair	↗ SM-24 + tab	↗ Click CH1+ screw terminal		6×0.22 cable core · cut 115 mm, twist the two cores ~1 turn/cm
6.2	Pair	↗ SM-24 – tab	↗ Click CH1– screw terminal		6×0.22 cable core · cut 105 mm, twist the two cores ~1 turn/cm
6.3	Top wire	AA18 GND rail	↗ Click CH1– screw terminal 2nd wire in CH1–: ties it to ground		insulated core · cut 55 mm, strip 5 mm (screw end) / 3 mm
6.4	Top wire	AA19 GND rail	↗ Click CH2– screw terminal grounds the unused channel		insulated core · cut 40 mm, strip 5 mm (screw end) / 3 mm
6.5	Top wire	↗ Click CH2+ screw terminal	↗ Click CH2– screw terminal short loop: CH2 shorted		insulated core · cut 20 mm, strip 5 mm (screw end) / 3 mm



COMPONENT SIDE · row A (back) at the top · column 1 on the left · solder-side wires shown through the board

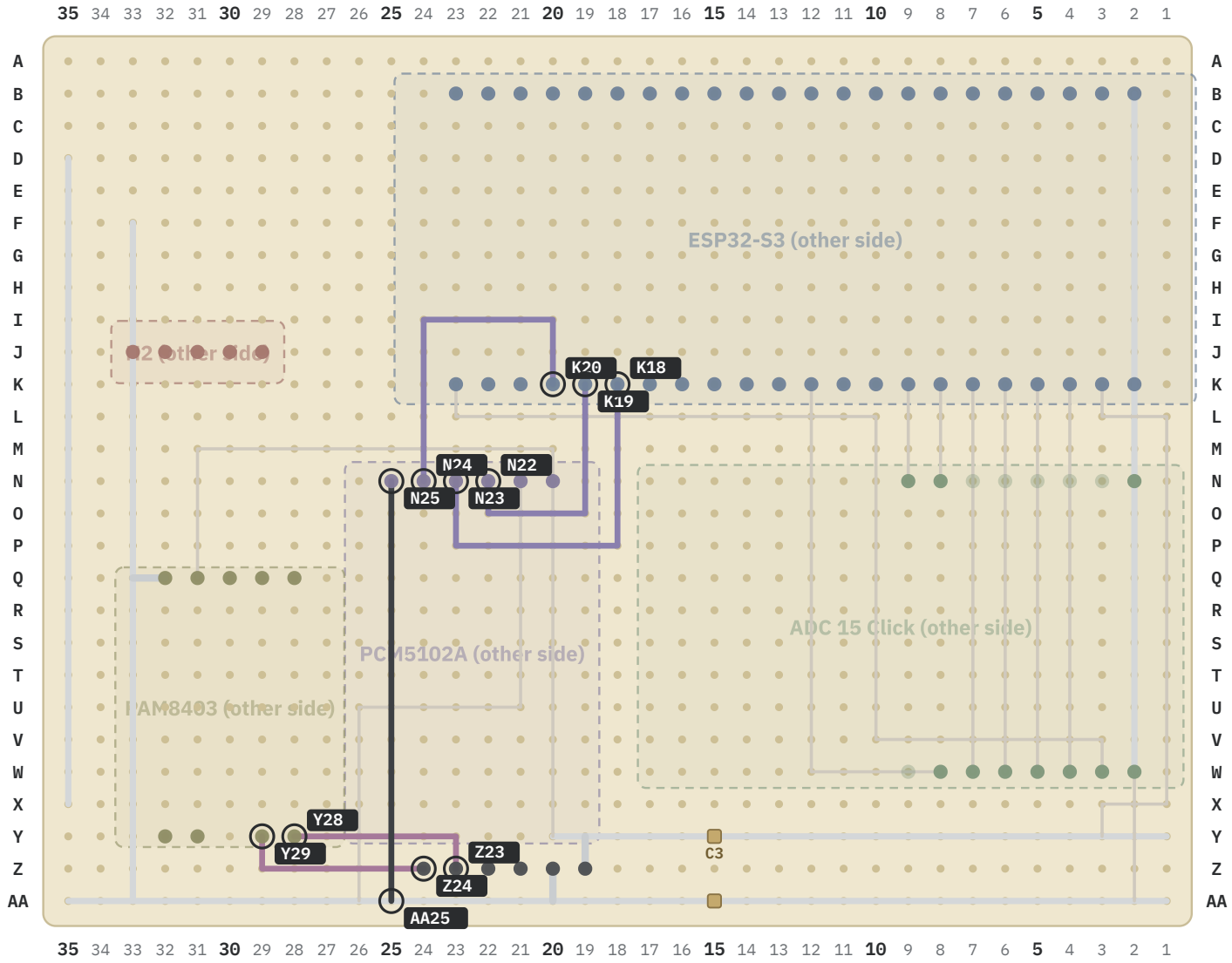
STEP 7 Audio: I²S, DAC → amp → H1

Soldering station

Before wiring, set the PCM5102A back-side jumpers: 1 FLT = L, 2 DEMP = L, 3 XSMT = H, 4 FMT = L. The two DAC → amp wires run on the component side, pad to pad on top of the modules.

DO THESE IN ORDER

#	TYPE	FROM	TO	ROUTE (SOLDER SIDE)	WIRE / CUT
7.1	Link	Y28 **PAM8403** L+ out	Z23 **H1** SPK+ speaker +	via Y23	insulated core · cut 25 mm, strip 3 mm both ends
7.2	Link	Y29 **PAM8403** L– out	Z24 **H1** SPK– speaker –, NOT ground	via Z29	insulated core · cut 25 mm, strip 3 mm both ends
7.3	Link	K20 **ESP32-S3** J1-4 GPIO4	N24 **PCM5102A** BCK bit clock	via I20 → I24	insulated core · cut 40 mm, strip 3 mm both ends
7.4	Link	K19 **ESP32-S3** J1-5 GPIO5	N22 **PCM5102A** LCK word clock	via O19 → O22	insulated core · cut 30 mm, strip 3 mm both ends
7.5	Link	K18 **ESP32-S3** J1-6 GPIO6	N23 **PCM5102A** DIN data	via P18 → P23	insulated core · cut 40 mm, strip 3 mm both ends
7.6	Link	N25 **PCM5102A** SCK	AA25 **GND rail** SCK → GND rail (SCK must be grounded)	straight	insulated core · cut 45 mm, strip 3 mm both ends
7.7	Top wire	↗ PCM5102A pad L (top)	↗ PAM8403 pad L in (top, above Q28)		insulated core · cut 40 mm, strip 5 mm (screw end) / 3 mm
7.8	Top wire	↗ PCM5102A pad AGND (top)	↗ PAM8403 pad ⏚ in (top, above Q29)		insulated core · cut 40 mm, strip 5 mm (screw end) / 3 mm



SOLDER SIDE · board turned over like a page · row A at the top · column 1 on the RIGHT

STEP 8 SIM868: power leads, UART, jumpers

Soldering station | Cutting bench

The Waveshare Pico-SIM868 board is used as a modem only: pull the Pico H off it, it is not used. Fit C1, then the four links on the solder side to H2. Then the two 0.5 mm² power leads from L35/L33 (top side) to a PH2.0 plug for the board's battery socket, and five M-F jumpers from H2 into the board's Pico sockets. Before power-on: GSM antenna on the "GSM" u.FL socket, nano SIM (PIN off) in the slot, the board's power slide switch on the battery side.

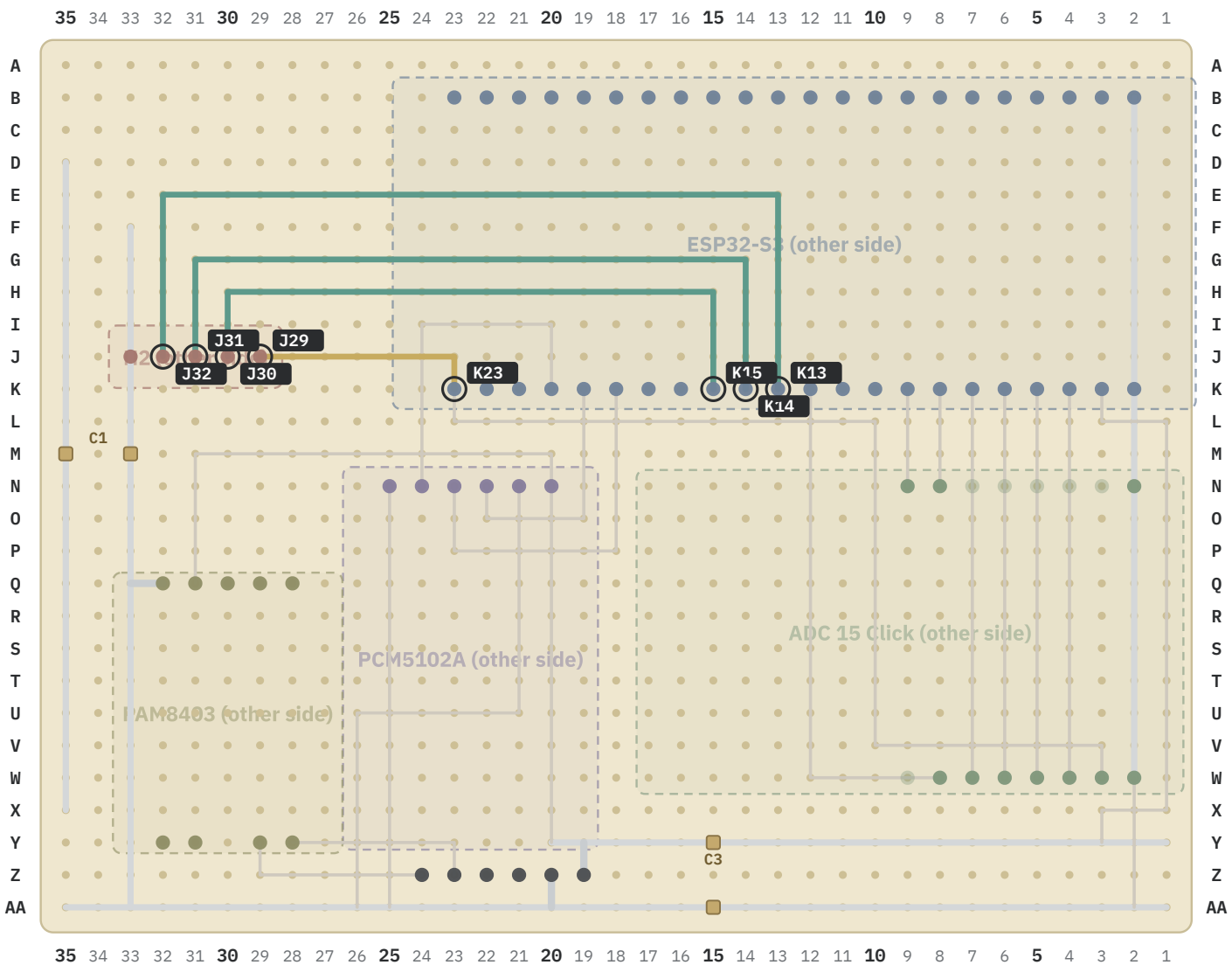
SIM868 BOARD: WHAT GOES WHERE

pin 1 (GP0)	SIM868 RXD, through the board's own 3.3 V ↔ 2.8 V level shifter. From H2 J31 ← GPIO17 (ESP TX). No divider needed.
pin 2 (GP1)	SIM868 TXD. To H2 J32 → GPIO18 (ESP RX).
pin 3	GND. From H2 J33 (GND bus col 33).
pin 19 (GP14)	PWR_EN: drives an NPN on the SIM868 PWRKEY. HIGH for 1.5 s switches the module on (or off). From H2 J30 ← GPIO16.
pin 36 (3V3 OUT)	Powers the board's level shifter, microphone and RTC backup. From H2 J29 ← ESP32 3V3. Without it the UART stays dead.
pin 39, pin 40	VSYS, VBUS: leave EMPTY. Never 5 V into this board.
BAT (PH2.0)	Battery socket → SIM868 VBAT (3.4–4.4 V, 2 A bursts). Red lead from L35 (VBAT rail), black from L33 (GND bus). Check polarity against the plug of the Li-po in the box.
GSM (u.FL)	GSM antenna, stuck flat to the inner wall at the back left. GNSS and BT sockets stay empty.

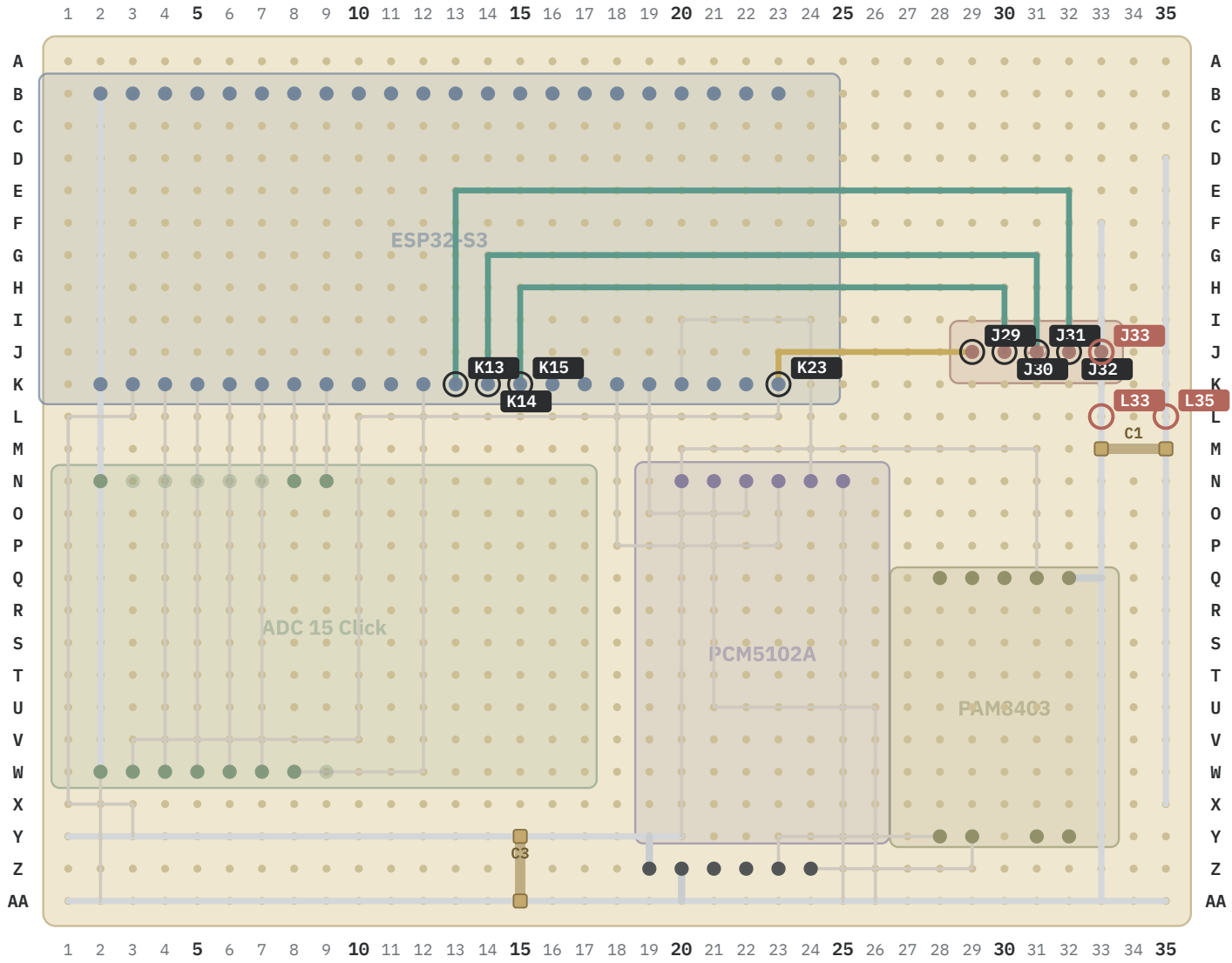
DO THESE IN ORDER

#	TYPE	FROM	TO	ROUTE (SOLDER SIDE)	WIRE / CUT
8.1	Part	**C1** 1000 µF · electrolytic, stripe = –, burst reserve at the SIM868 power take-off legs into M35 + M33 – (component side)			bend legs flat after soldering
8.2	Link	K23 **ESP32-S3** J1-1 3V3	J29 **H2** 3V3 → SIM868 pin 36 ESP32 3V3 (J1-1) → H2 3V3 (feeds the SIM868 level shifter)	via J23	insulated core · cut 30 mm, strip 3 mm both ends
8.3	Link	K15 **ESP32-S3** J1-9 GPIO16	J30 **H2** PWRKEY → SIM868 pin 19 GPIO16 → H2 PWRKEY (HIGH 1.5 s = power on/off)	via H15 → H30	insulated core · cut 60 mm, strip 3 mm both ends
8.4	Link	K14 **ESP32-S3** J1-10 GPIO17	J31 **H2** SIM RXD → SIM868 pin 1 GPIO17 (ESP TX) → H2 SIM RXD	via G14 → G31	insulated core · cut 70 mm, strip 3 mm both ends
8.5	Link	J32 **H2** SIM TXD → SIM868 pin 2	K13 **ESP32-S3** J1-11 GPIO18 H2 SIM TXD → GPIO18 (ESP RX)	via E32 → E13	insulated core · cut 85 mm, strip 3 mm both ends
8.6	0.5 mm²	L35 VBAT rail, next to C1+	↗ SIM868 battery socket + (PH2.0 plug, red lead) SIM868 power: straight from the switched 18650, never 5 V		0.5 mm² red · cut 80 mm, strip 4 mm, tin
8.7	0.5 mm²	L33 GND bus col 33, next to C1–	↗ SIM868 battery socket – (PH2.0 plug, black lead)		0.5 mm² black · cut 75 mm, strip 4 mm, tin
8.8	Jumper	J29 H2 3V3	↗ SIM868 header pin 36 (3V3(OUT)) female end on H2, male end straight into the SIM868 socket (Pico removed)		M-F jumper 10 cm (stock row 39), used as is: female end on H2, male end into the SIM868 socket
8.9	Jumper	J30 H2 PWRKEY	↗ SIM868 header pin 19 (GP14 / PWR_EN) female end on H2, male end straight into the SIM868 socket (Pico removed)		M-F jumper 10 cm (stock row 39), used as is: female end on H2, male end into the SIM868 socket
8.10	Jumper	J31 H2 SIM RXD	↗ SIM868 header pin 1 (GP0 / RXD) female end on H2, male end straight into the SIM868 socket (Pico removed)		M-F jumper 10 cm (stock row 39), used as is: female end on H2, male end into the SIM868 socket

#	TYPE	FROM	TO	ROUTE (SOLDER SIDE)	WIRE / CUT
8.11	Jumper	J32 H2 SIM TXD	↗ SIM868 header pin 2 (GP1 / TXD) female end on H2, male end straight into the SIM868 socket (Pico removed)		M-F jumper 10 cm (stock row 39), used as is: female end on H2, male end into the SIM868 socket
8.12	Jumper	J33 H2 GND	↗ SIM868 header pin 3 (GND) female end on H2, male end straight into the SIM868 socket (Pico removed)		M-F jumper 10 cm (stock row 39), used as is: female end on H2, male end into the SIM868 socket



SOLDER SIDE · board turned over like a page · row A at the top · column 1 on the RIGHT



COMPONENT SIDE · row A (back) at the top · column 1 on the left · solder-side wires shown through the board

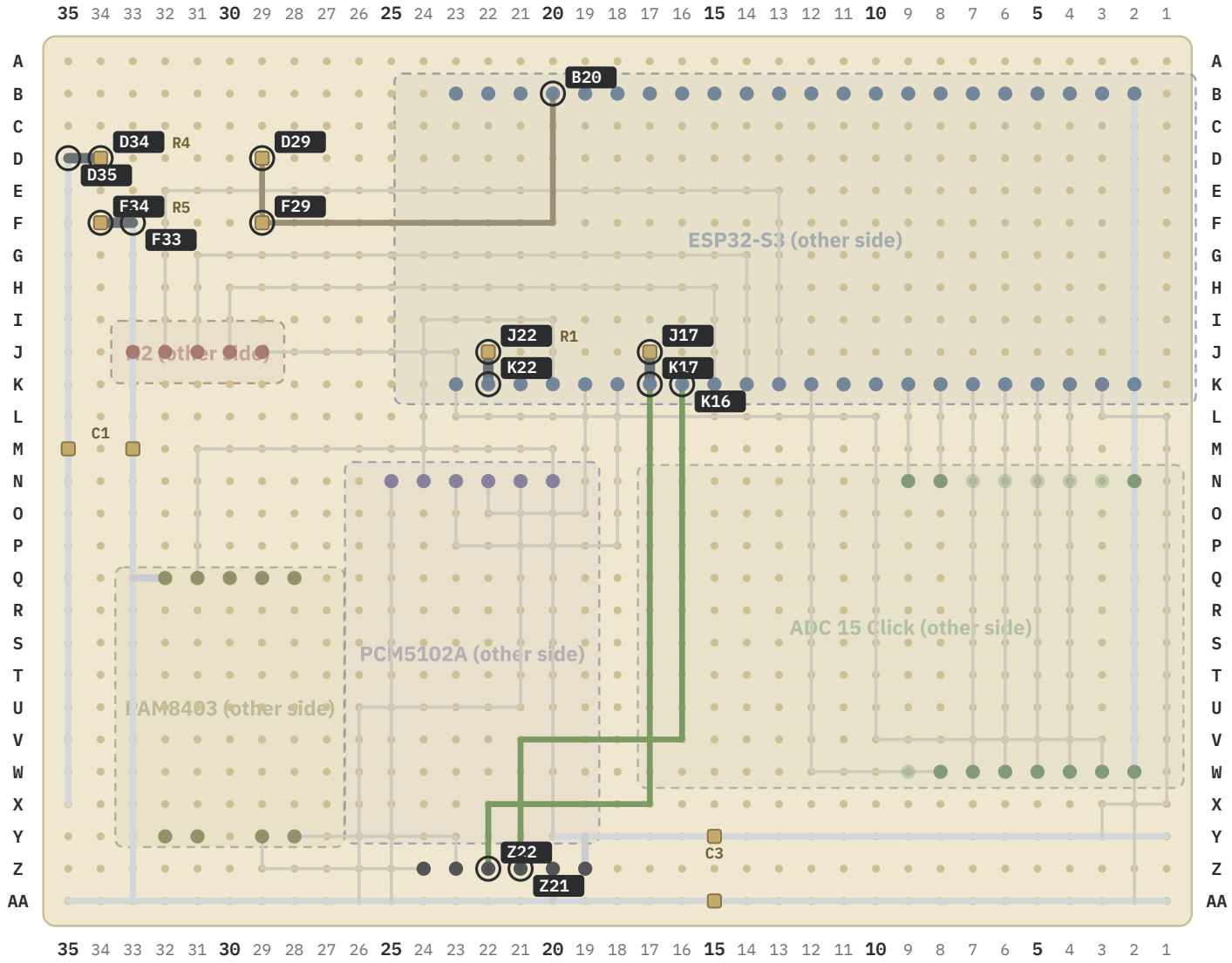
STEP 9 Battery sense, button & LED lines

Soldering station

R4, R5 (free area at the back right, rows D and F) and R1 (under the ESP32-S3) go in from the component side; bridges and links on the solder side.

DO THESE IN ORDER

#	TYPE	FROM	TO	ROUTE (SOLDER SIDE)	WIRE / CUT
9.1	Part	**R1** 10 kΩ · button pull-up legs into J17 J22 (component side)			bend legs flat after soldering
9.2	Part	**R4** 100 kΩ · battery divider, top legs into D34 D29 (component side)			bend legs flat after soldering
9.3	Part	**R5** 100 kΩ · battery divider, bottom legs into F29 F34 (component side)			bend legs flat after soldering
9.4	Bridge	J17 **R1** 10 kΩ leg	K17 **ESP32-S3** J1-7 GPIO7 R1 leg → GPIO7		solder bridge / bent leg
9.5	Bridge	J22 **R1** 10 kΩ leg	K22 **ESP32-S3** J1-2 3V3 R1 leg → 3V3 (J1-2)		solder bridge / bent leg
9.6	Bridge	D34 **R4** 100 kΩ leg	D35 **VBAT rail** R4 leg → VBAT rail		solder bridge / bent leg
9.7	Bridge	F34 **R5** 100 kΩ leg	F33 **GND bus** col 33 R5 leg → GND bus col 33		solder bridge / bent leg
9.8	Link	K17 **ESP32-S3** J1-7 GPIO7	Z22 **H1** BTN button line	via X17 → X22	insulated core · cut 60 mm, strip 3 mm both ends
9.9	Link	K16 **ESP32-S3** J1-8 GPIO15	Z21 **H1** LED LED data	via V16 → V21	insulated core · cut 60 mm, strip 3 mm both ends
9.10	Link	D29 **R4** 100 kΩ leg	F29 **R5** 100 kΩ leg divider node R4–R5	straight	insulated core · cut 15 mm, strip 3 mm both ends
9.11	Link	F29 **R5** 100 kΩ leg	B20 **ESP32-S3** J3-4 GPIO1 VBAT ÷ 2 → GPIO1	via F20	insulated core · cut 45 mm, strip 3 mm both ends



SOLDER SIDE · board turned over like a page · row A at the top · column 1 on the RIGHT

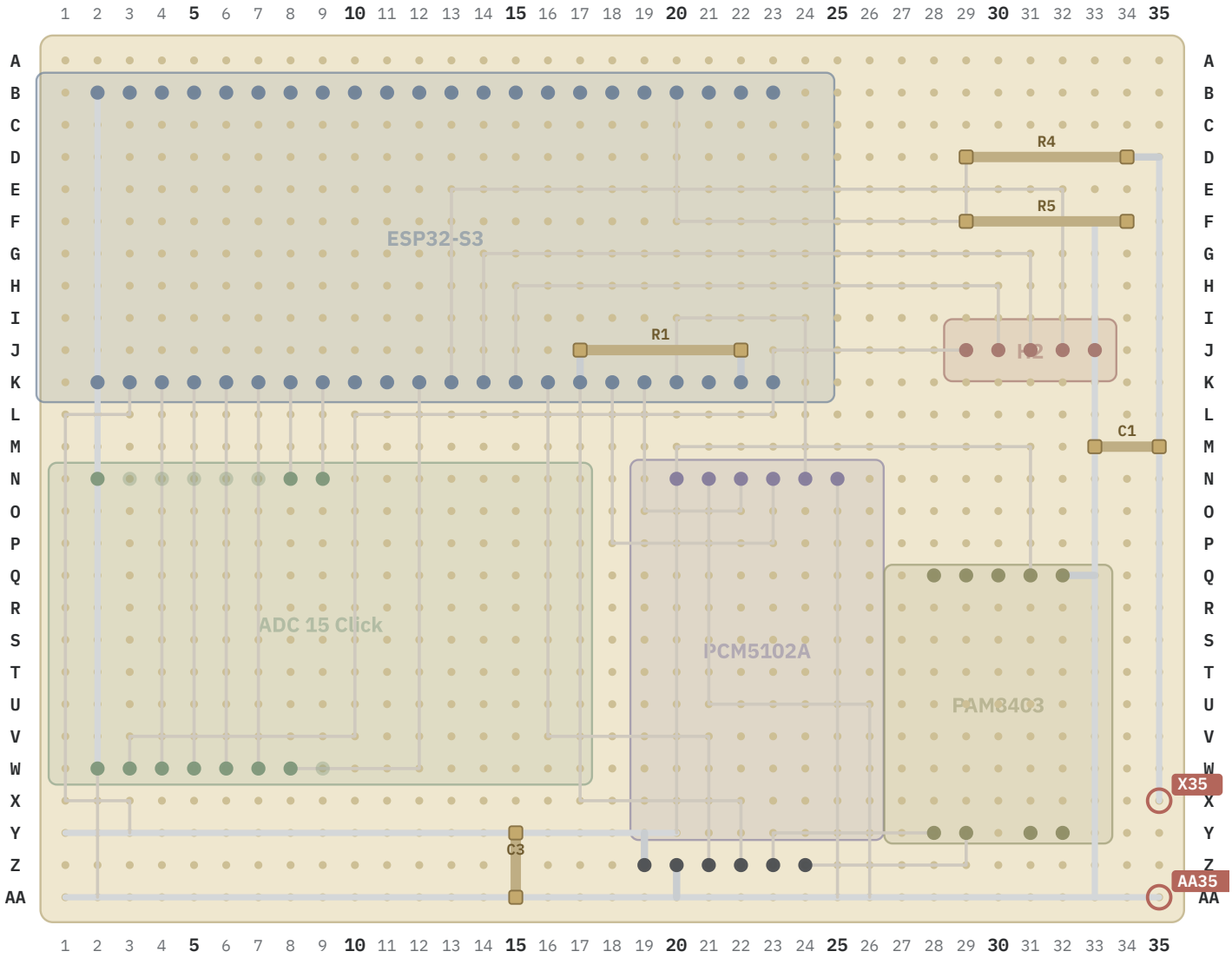
STEP 10 Battery → charger → switch

Soldering station

Battery out of the holder. Push the rocker into the base's wall cutout from outside first, then solder its two pins (lamp pin unused). Rail-end holes: tin the 0.5 mm² wire, push it through the hole from the component side and solder it to the rail underneath.

DO THESE IN ORDER

#	TYPE	FROM	TO	ROUTE (SOLDER SIDE)	WIRE / CUT
10.1	Lead	↗ Holder + lead	↗ TP4056 B+		holder lead · trim to 40 mm, strip 4 mm
10.2	Lead	↗ Holder – lead	↗ TP4056 B–		holder lead · trim to 40 mm, strip 4 mm
10.3	0.5 mm²	↗ TP4056 OUT+	↗ Rocker pin 1		0.5 mm² red · cut 55 mm, strip 4 mm, tin
10.4	0.5 mm²	↗ Rocker pin 2	X35 VBAT rail end		0.5 mm² red · cut 60 mm, strip 4 mm, tin
10.5	0.5 mm²	↗ TP4056 OUT–	AA35 GND rail end		0.5 mm² black · cut 55 mm, strip 4 mm, tin



COMPONENT SIDE · row A (back) at the top · column 1 on the left · solder-side wires shown through the board

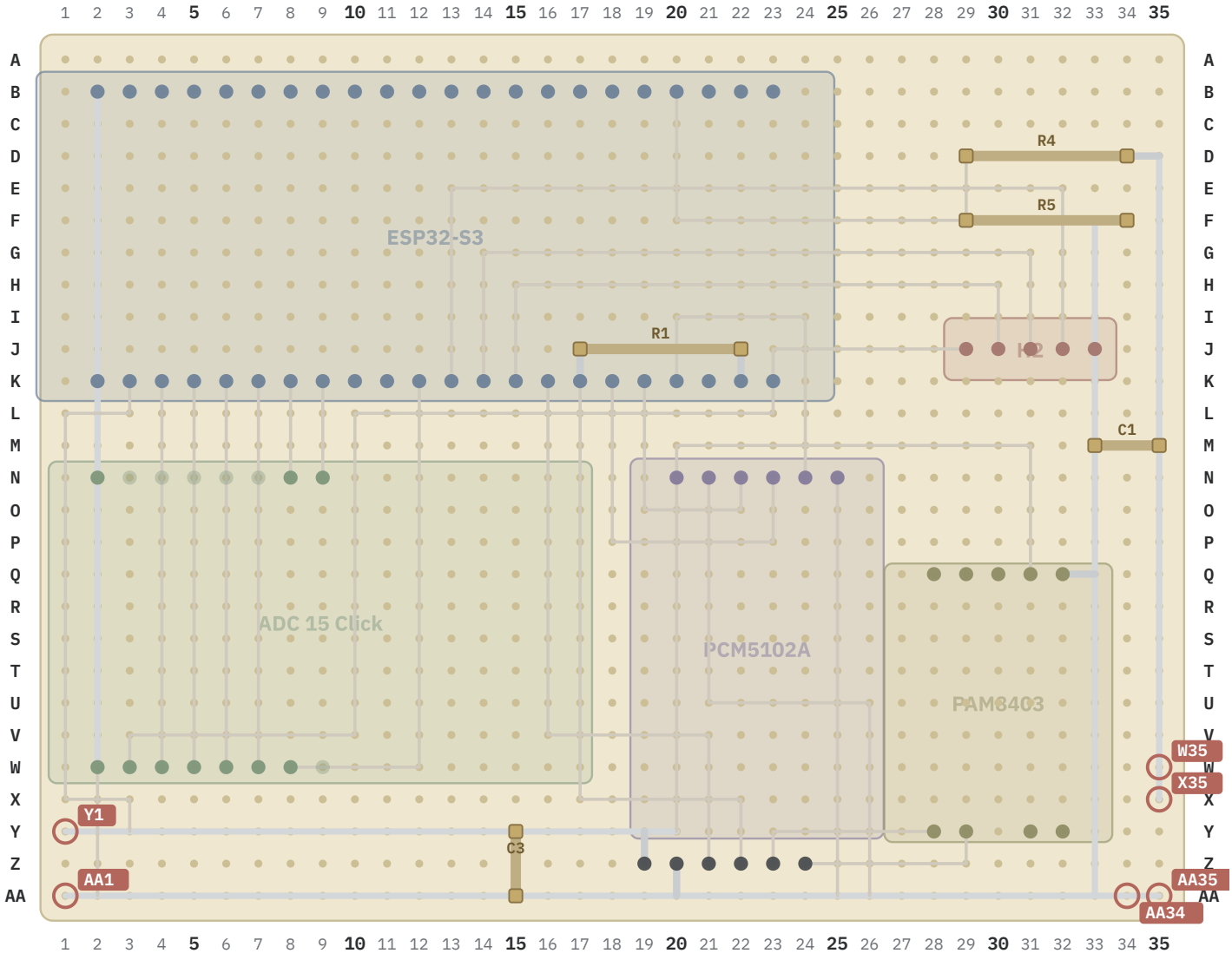
STEP 11 Boost converter to 5.0 V

Multimeter · Soldering station

Before connecting the output: power the boost input from the battery, turn its trimmer until the output reads 5.00 V (±0.05 V). Then wire it.

DO THESE IN ORDER

#	TYPE	FROM	TO	ROUTE (SOLDER SIDE)	WIRE / CUT
11.1	0.5 mm²	W35 VBAT rail	↗ Boost IN+		0.5 mm² red · cut 90 mm, strip 4 mm, tin
11.2	0.5 mm²	AA34 GND rail	↗ Boost IN−		0.5 mm² black · cut 95 mm, strip 4 mm, tin
11.3	0.5 mm²	↗ Boost OUT+	Y1 5 V rail start		0.5 mm² red · cut 120 mm, strip 4 mm, tin
11.4	0.5 mm²	↗ Boost OUT−	AA1 GND rail start		0.5 mm² black · cut 125 mm, strip 4 mm, tin



COMPONENT SIDE · row A (back) at the top · column 1 on the left · solder-side wires shown through the board

STEP 12 Lid board: button, LED, diode

Soldering station · Cutting bench

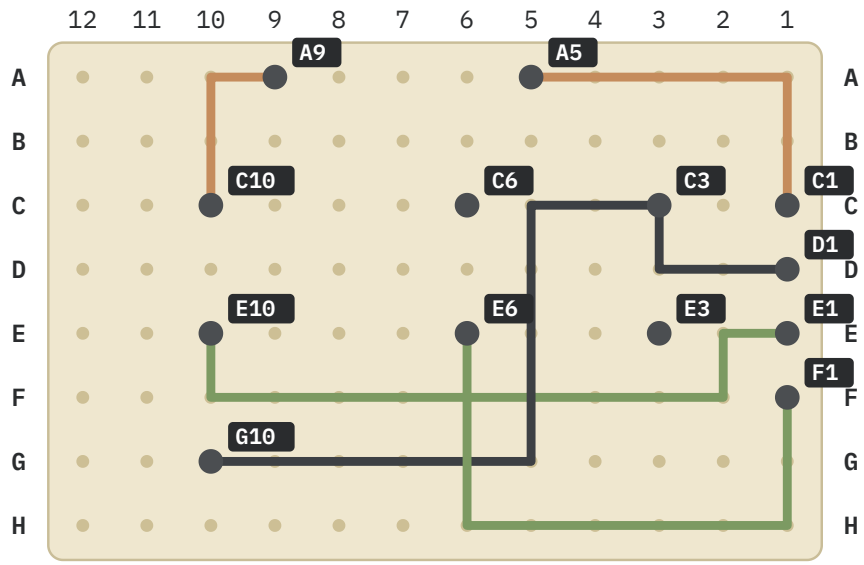
Lid board, component side up: fit the parts into the listed holes, then turn it over and make the links. The header pins are bent 90° outward over the edge after soldering.

PARTS INTO HOLES (COMPONENT SIDE)

C1 D1 E1 F1	Lid header, 4 straight pins (C1 5V, D1 GND, E1 DIN, F1 BTN), bent 90° outward over the edge
C3 C6 E3 E6	Tact switch legs (C3 + C6 are joined inside, E3 + E6 are joined inside)
A5 → A9	1N4001: anode A5, cathode (stripe) A9
C10 E10 G10	WS2812B single pixel: stubs (resistor-leg offcuts) from its +5V, DIN, GND pads into C10, E10, G10

DO THESE IN ORDER

#	TYPE	FROM	TO	ROUTE (SOLDER SIDE)	WIRE / CUT
12.1	Link	C1 **Lid header** 5V	A5 **1N4001** anode header 5V → diode anode	via A1	insulated core · cut 25 mm, strip 3 mm both ends
12.2	Link	A9 **1N4001** cathode (stripe)	C10 **WS2812B** +5V stub diode cathode → pixel +5V	via A10	insulated core · cut 20 mm, strip 3 mm both ends
12.3	Link	E1 **Lid header** DIN	E10 **WS2812B** DIN stub header DIN → pixel DIN	via E2 → F2 → F10	insulated core · cut 40 mm, strip 3 mm both ends
12.4	Link	D1 **Lid header** GND	C3 **Button** leg 1 header GND → button leg 1	via D3	insulated core · cut 20 mm, strip 3 mm both ends
12.5	Link	C3 **Button** leg 1	G10 **WS2812B** GND stub button leg 1 → pixel GND	via C5 → G5	insulated core · cut 40 mm, strip 3 mm both ends
12.6	Link	F1 **Lid header** BTN	E6 **Button** leg 4 header BTN → button leg 4 (diagonal)	via H1 → H6	insulated core · cut 35 mm, strip 3 mm both ends



LID BOARD · LINK SIDE (turned over like a page: column 1 on the RIGHT)

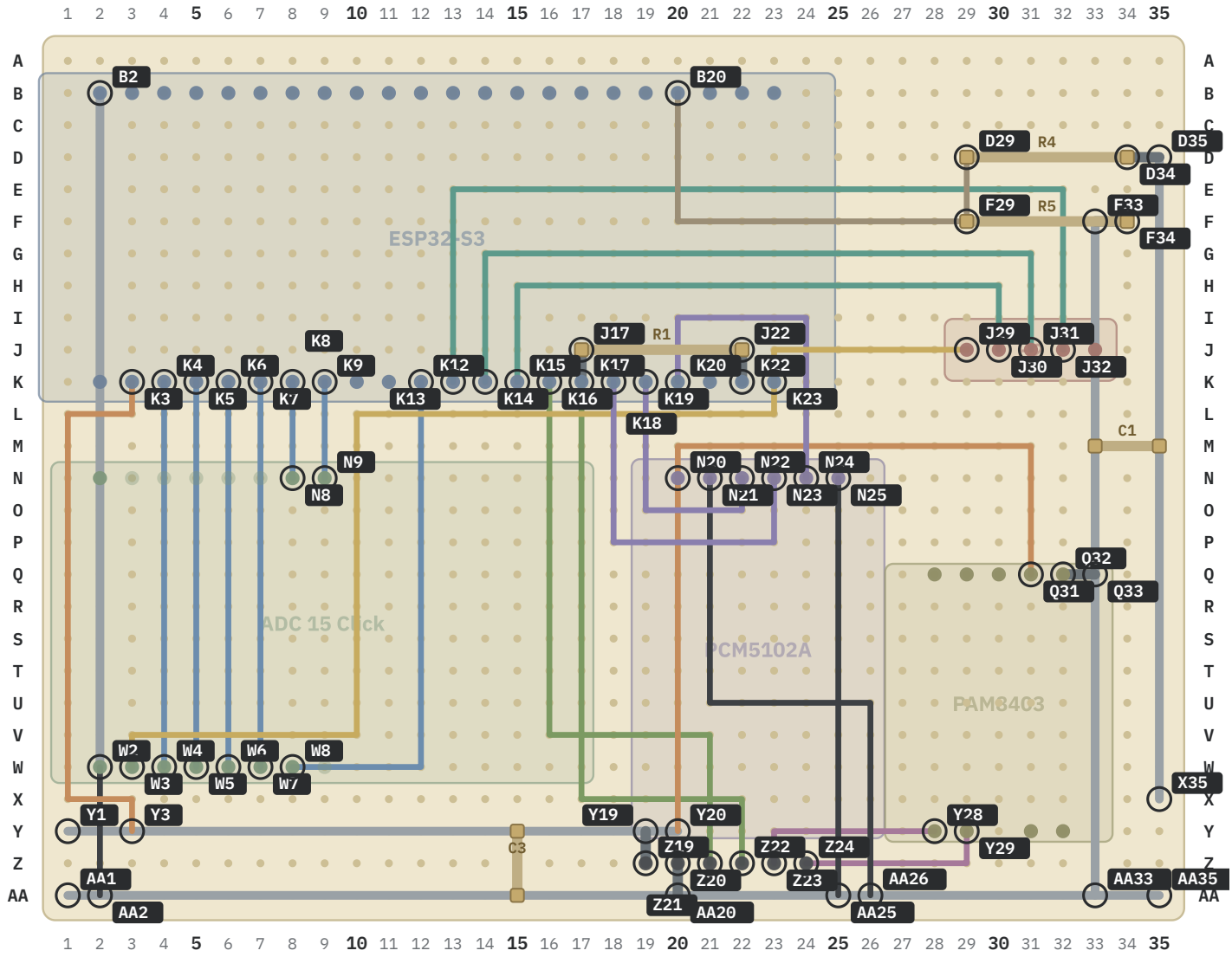
STEP 13 Lid harness & speaker leads

Cutting bench · Soldering station

Four F-F jumpers are shortened (cut the stated length out of the middle, strip 4 mm, twist, solder, heat-shrink) and two are cut for the speaker (keep one female plug, solder the cut end to the speaker tab, heat-shrink).

DO THESE IN ORDER

#	TYPE	FROM	TO	ROUTE (SOLDER SIDE)	WIRE / CUT
13.1	Jumper	H1 5V Z19	↗ Lid header 5V		F-F jumper → 60 mm: cut 40 mm out of the middle, splice + heat-shrink
13.2	Jumper	H1 GND Z20	↗ Lid header GND		F-F jumper → 60 mm: cut 40 mm out of the middle, splice + heat-shrink
13.3	Jumper	H1 LED Z21	↗ Lid header DIN		F-F jumper → 60 mm: cut 40 mm out of the middle, splice + heat-shrink
13.4	Jumper	H1 BTN Z22	↗ Lid header BTN		F-F jumper → 60 mm: cut 40 mm out of the middle, splice + heat-shrink
13.5	Jumper	H1 SPK+ Z23	↗ Speaker + tab		F-F jumper → 75 mm: keep one female end, cut 25 mm off, solder to the speaker tab
13.6	Jumper	H1 SPK– Z24	↗ Speaker – tab		F-F jumper → 90 mm: keep one female end, cut 10 mm off, solder to the speaker tab



COMPONENT SIDE · row A (back) at the top · column 1 on the left · solder-side wires shown through the board

STEP 14 Check sheet

Battery out, all modules unplugged. Beep each pair with the meter. Tick when done.

MUST NOT BEEP (SHORTS)

☐	AA1 ↔ Y1	GND ↔ 5 V
☐	AA1 ↔ X35	GND ↔ VBAT
☐	AA1 ↔ K23	GND ↔ 3V3
☐	Y1 ↔ X35	5 V ↔ VBAT
☐	Y1 ↔ L35	5 V ↔ SIM868 VBAT lead (would destroy the SIM868)
☐	Y1 ↔ J29	5 V ↔ H2 3V3
☐	L35 ↔ J29	VBAT ↔ H2 3V3
☐	Y28 ↔ AA1	PAM L+ ↔ GND
☐	Y29 ↔ AA1	PAM L− ↔ GND

MUST BEEP (EVERY LINK, END TO END)

☐	Z19 ↔ Y19	H1 5V → 5 V rail	step 4
☐	Z20 ↔ AA20	H1 GND → GND rail	step 4
☐	Q32 ↔ Q33	PAM8403 GND → GND bus	step 4
☐	J17 ↔ K17	R1 leg → GPIO7	step 9
☐	J22 ↔ K22	R1 leg → 3V3 (J1-2)	step 9
☐	D34 ↔ D35	R4 leg → VBAT rail	step 9
☐	F34 ↔ F33	R5 leg → GND bus col 33	step 9
☐	K6 ↔ W6	SCK	step 5
☐	K5 ↔ W5	MISO	step 5
☐	K4 ↔ W4	MOSI	step 5
☐	K7 ↔ W7	ADC chip-select	step 5
☐	K9 ↔ N9	clock-chip select	step 5
☐	K8 ↔ N8	DRDY, active low	step 5
☐	K12 ↔ W8	ADC reset	step 5
☐	K17 ↔ Z22	button line	step 9
☐	K16 ↔ Z21	LED data	step 9
☐	Y28 ↔ Z23	speaker +	step 7
☐	Y29 ↔ Z24	speaker −, NOT ground	step 7
☐	W2 ↔ AA2	GND bus col 2 → GND rail	step 4
☐	Y3 ↔ K3	5 V rail → ESP32 5V	step 4
☐	Y20 ↔ N20	5 V rail → PCM VIN	step 4
☐	N20 ↔ Q31	PCM VIN → PAM8403 +5V	step 4
☐	N21 ↔ AA26	PCM GND → GND rail	step 4
☐	K23 ↔ W3	ESP32 3V3 → Click 3V3	step 4
☐	K20 ↔ N24	bit clock	step 7
☐	K19 ↔ N22	word clock	step 7
☐	K18 ↔ N23	data	step 7
☐	N25 ↔ AA25	SCK → GND rail (SCK must be grounded)	step 7
☐	K23 ↔ J29	ESP32 3V3 (J1-1) → H2 3V3 (feeds the SIM868 level shifter)	step 8
☐	K15 ↔ J30	GPIO16 → H2 PWRKEY (HIGH 1.5 s = power on/off)	step 8
☐	K14 ↔ J31	GPIO17 (ESP TX) → H2 SIM RXD	step 8
☐	J32 ↔ K13	H2 SIM TXD → GPIO18 (ESP RX)	step 8
☐	D29 ↔ F29	divider node R4–R5	step 9

☐	`F29` ↔ `B20`	VBAT ÷ 2 → GPIO1	step 9
☐	`C1` ↔ `A5` (lid)	header 5V → diode anode	step 12
☐	`A9` ↔ `C10` (lid)	diode cathode → pixel +5V	step 12
☐	`E1` ↔ `E10` (lid)	header DIN → pixel DIN	step 12
☐	`D1` ↔ `C3` (lid)	header GND → button leg 1	step 12
☐	`C3` ↔ `G10` (lid)	button leg 1 → pixel GND	step 12
☐	`F1` ↔ `E6` (lid)	header BTN → button leg 4 (diagonal)	step 12

THEN, MODULES PLUGGED IN, BATTERY IN, ROCKER ON

- ☐ 5.0 V between Z19 and Z20 (H1 5V–GND)
- ☐ 3.3 V between W3 and W2 (Click 3V3–GND)
- ☐ 3.3 V between J29 and J33 (H2 3V3–GND)
- ☐ 3.7–4.2 V between L35 and L33 (SIM868 battery lead) and the same on the PH2.0 plug, red = +
- ☐ SIM868: PWRKEY pulse → NET LED blinks fast, then about every 3 s (registered); AT → AT+CPIN? → AT+CREG? (0,1) → test call
- ☐ Rocker OFF before plugging USB-C into the ESP32-S3 to flash

CUT Cut list (every wire cut to length)

STEP	TYPE	FROM	TO	LENGTH	INSTRUCTION
4	Bare wire	AA1 **GND rail**	AA35 **GND rail**	**100 mm**	bare 20 AWG · cut 100 mm
4	Bare wire	Y1 **5 V rail**	Y20 **5 V rail**	**60 mm**	bare 20 AWG · cut 60 mm
4	Bare wire	B2 **ESP32-S3** J3-22 GND	W2 **Click** GND	**65 mm**	bare 20 AWG · cut 65 mm
4	Bare wire	F33 **GND bus** col 33	AA33 **GND rail**	**65 mm**	bare 20 AWG · cut 65 mm
4	Bare wire	D35 **VBAT rail**	X35 **VBAT rail**	**65 mm**	bare 20 AWG · cut 65 mm
4	Link	W2 **Click** GND	AA2 **GND rail**	**20 mm**	insulated core · cut 20 mm, strip 3 mm both ends
4	Link	Y3 **5 V rail**	K3 **ESP32-S3** J1-21 5V	**55 mm**	insulated core · cut 55 mm, strip 3 mm both ends
4	Link	Y20 **5 V rail**	N20 **PCM5102A** VIN	**40 mm**	insulated core · cut 40 mm, strip 3 mm both ends
4	Link	N20 **PCM5102A** VIN	Q31 **PAM8403** +5V	**50 mm**	insulated core · cut 50 mm, strip 3 mm both ends
4	Link	N21 **PCM5102A** GND	AA26 **GND rail**	**55 mm**	insulated core · cut 55 mm, strip 3 mm both ends
4	Link	K23 **ESP32-S3** J1-1 3V3	W3 **Click** 3V3	**90 mm**	insulated core · cut 90 mm, strip 3 mm both ends
5	Link	K6 **ESP32-S3** J1-18 GPIO12	W6 **Click** SCK	**40 mm**	insulated core · cut 40 mm, strip 3 mm both ends
5	Link	K5 **ESP32-S3** J1-19 GPIO13	W5 **Click** SDO (MISO)	**40 mm**	insulated core · cut 40 mm, strip 3 mm both ends
5	Link	K4 **ESP32-S3** J1-20 GPIO14	W4 **Click** SDI (MOSI)	**40 mm**	insulated core · cut 40 mm, strip 3 mm both ends
5	Link	K7 **ESP32-S3** J1-17 GPIO11	W7 **Click** CS1	**40 mm**	insulated core · cut 40 mm, strip 3 mm both ends
5	Link	K9 **ESP32-S3** J1-15 GPIO9	N9 **Click** CS2	**20 mm**	insulated core · cut 20 mm, strip 3 mm both ends
5	Link	K8 **ESP32-S3** J1-16 GPIO10	N8 **Click** INT (DRDY)	**20 mm**	insulated core · cut 20 mm, strip 3 mm both ends
5	Link	K12 **ESP32-S3** J1-12 GPIO8	W8 **Click** RST	**50 mm**	insulated core · cut 50 mm, strip 3 mm both ends
6	Pair	↗ SM-24 + tab	↗ Click CH1+ screw terminal	**115 mm**	6×0.22 cable core · cut 115 mm, twist the two cores ~1 turn/cm
6	Pair	↗ SM-24 – tab	↗ Click CH1– screw terminal	**105 mm**	6×0.22 cable core · cut 105 mm, twist the two cores ~1 turn/cm
6	Top wire	AA18 GND rail	↗ Click CH1– screw terminal	**55 mm**	insulated core · cut 55 mm, strip 5 mm (screw end) / 3 mm
6	Top wire	AA19 GND rail	↗ Click CH2– screw terminal	**40 mm**	insulated core · cut 40 mm, strip 5 mm (screw end) / 3 mm
6	Top wire	↗ Click CH2+ screw terminal	↗ Click CH2– screw terminal	**20 mm**	insulated core · cut 20 mm, strip 5 mm (screw end) / 3 mm
7	Link	Y28 **PAM8403** L+ out	Z23 **H1** SPK+	**25 mm**	insulated core · cut 25 mm, strip 3 mm both ends
7	Link	Y29 **PAM8403** L– out	Z24 **H1** SPK–	**25 mm**	insulated core · cut 25 mm, strip 3 mm both ends
7	Link	K20 **ESP32-S3** J1-4 GPIO4	N24 **PCM5102A** BCK	**40 mm**	insulated core · cut 40 mm, strip 3 mm both ends
7	Link	K19 **ESP32-S3** J1-5 GPIO5	N22 **PCM5102A** LCK	**30 mm**	insulated core · cut 30 mm, strip 3 mm both ends

STEP	TYPE	FROM	TO	LENGTH	INSTRUCTION
7	Link	K18 **ESP32-S3** J1-6 GPIO6	N23 **PCM5102A** DIN	**40 mm**	insulated core · cut 40 mm, strip 3 mm both ends
7	Link	N25 **PCM5102A** SCK	AA25 **GND rail**	**45 mm**	insulated core · cut 45 mm, strip 3 mm both ends
7	Top wire	↗ PCM5102A pad L (top)	↗ PAM8403 pad L in (top, above Q28)	**40 mm**	insulated core · cut 40 mm, strip 5 mm (screw end) / 3 mm
7	Top wire	↗ PCM5102A pad AGND (top)	↗ PAM8403 pad ⏚ in (top, above Q29)	**40 mm**	insulated core · cut 40 mm, strip 5 mm (screw end) / 3 mm
8	Link	K23 **ESP32-S3** J1-1 3V3	J29 **H2** 3V3 → SIM868 pin 36	**30 mm**	insulated core · cut 30 mm, strip 3 mm both ends
8	Link	K15 **ESP32-S3** J1-9 GPIO16	J30 **H2** PWRKEY → SIM868 pin 19	**60 mm**	insulated core · cut 60 mm, strip 3 mm both ends
8	Link	K14 **ESP32-S3** J1-10 GPIO17	J31 **H2** SIM RXD → SIM868 pin 1	**70 mm**	insulated core · cut 70 mm, strip 3 mm both ends
8	Link	J32 **H2** SIM TXD → SIM868 pin 2	K13 **ESP32-S3** J1-11 GPIO18	**85 mm**	insulated core · cut 85 mm, strip 3 mm both ends
8	0.5 mm²	L35 VBAT rail, next to C1+	↗ SIM868 battery socket + (PH2.0 plug, red lead)	**5 mm**	0.5 mm² red · cut 80 mm, strip 4 mm, tin
8	0.5 mm²	L33 GND bus col 33, next to C1−	↗ SIM868 battery socket − (PH2.0 plug, black lead)	**5 mm**	0.5 mm² black · cut 75 mm, strip 4 mm, tin
9	Link	K17 **ESP32-S3** J1-7 GPIO7	Z22 **H1** BTN	**60 mm**	insulated core · cut 60 mm, strip 3 mm both ends
9	Link	K16 **ESP32-S3** J1-8 GPIO15	Z21 **H1** LED	**60 mm**	insulated core · cut 60 mm, strip 3 mm both ends
9	Link	D29 **R4** 100 kΩ leg	F29 **R5** 100 kΩ leg	**15 mm**	insulated core · cut 15 mm, strip 3 mm both ends
9	Link	F29 **R5** 100 kΩ leg	B20 **ESP32-S3** J3-4 GPIO1	**45 mm**	insulated core · cut 45 mm, strip 3 mm both ends
10	Lead	↗ Holder + lead	↗ TP4056 B+	**40 mm**	holder lead · trim to 40 mm, strip 4 mm
10	Lead	↗ Holder − lead	↗ TP4056 B−	**40 mm**	holder lead · trim to 40 mm, strip 4 mm
10	0.5 mm²	↗ TP4056 OUT+	↗ Rocker pin 1	**5 mm**	0.5 mm² red · cut 55 mm, strip 4 mm, tin
10	0.5 mm²	↗ Rocker pin 2	X35 VBAT rail end	**5 mm**	0.5 mm² red · cut 60 mm, strip 4 mm, tin
10	0.5 mm²	↗ TP4056 OUT−	AA35 GND rail end	**5 mm**	0.5 mm² black · cut 55 mm, strip 4 mm, tin
11	0.5 mm²	W35 VBAT rail	↗ Boost IN+	**5 mm**	0.5 mm² red · cut 90 mm, strip 4 mm, tin
11	0.5 mm²	AA34 GND rail	↗ Boost IN−	**5 mm**	0.5 mm² black · cut 95 mm, strip 4 mm, tin
11	0.5 mm²	↗ Boost OUT+	Y1 5 V rail start	**5 mm**	0.5 mm² red · cut 120 mm, strip 4 mm, tin
11	0.5 mm²	↗ Boost OUT−	AA1 GND rail start	**5 mm**	0.5 mm² black · cut 125 mm, strip 4 mm, tin
12	Link	C1 **Lid header** 5V	A5 **1N4001** anode	**25 mm**	insulated core · cut 25 mm, strip 3 mm both ends
12	Link	A9 **1N4001** cathode (stripe)	C10 **WS2812B** +5V stub	**20 mm**	insulated core · cut 20 mm, strip 3 mm both ends
12	Link	E1 **Lid header** DIN	E10 **WS2812B** DIN stub	**40 mm**	insulated core · cut 40 mm, strip 3 mm both ends
12	Link	D1 **Lid header** GND	C3 **Button** leg 1	**20 mm**	insulated core · cut 20 mm, strip 3 mm both ends

STEP	TYPE	FROM	TO	LENGTH	INSTRUCTION
12	● **Link**	**C3** **Button** leg 1	**G10** **WS2812B** GND stub	**40 mm**	insulated core · cut 40 mm, strip 3 mm both ends
12	● **Link**	**F1** **Lid header** BTN	**E6** **Button** leg 4	**35 mm**	insulated core · cut 35 mm, strip 3 mm both ends
13	● **Jumper**	**H1** 5V Z19	↗ Lid header 5V	**60 mm**	F-F jumper → 60 mm: cut 40 mm out of the middle, splice + heat-shrink
13	● **Jumper**	**H1** GND Z20	↗ Lid header GND	**60 mm**	F-F jumper → 60 mm: cut 40 mm out of the middle, splice + heat-shrink
13	● **Jumper**	**H1** LED Z21	↗ Lid header DIN	**60 mm**	F-F jumper → 60 mm: cut 40 mm out of the middle, splice + heat-shrink
13	● **Jumper**	**H1** BTN Z22	↗ Lid header BTN	**60 mm**	F-F jumper → 60 mm: cut 40 mm out of the middle, splice + heat-shrink
13	● **Jumper**	**H1** SPK+ Z23	↗ Speaker + tab	**75 mm**	F-F jumper → 75 mm: keep one female end, cut 25 mm off, solder to the speaker tab
13	● **Jumper**	**H1** SPK− Z24	↗ Speaker − tab	**90 mm**	F-F jumper → 90 mm: keep one female end, cut 10 mm off, solder to the speaker tab

Totals: insulated cores ≈ 1370 mm (≈ 280 mm of the 6-core cable), bare 20 AWG ≈ 355 mm. Module outlines on the maps are approximate; hole positions are exact.